Arts+STEM Hub Seminar · Mary Immaculate College · 4 November 2026

From Ideas to Impact

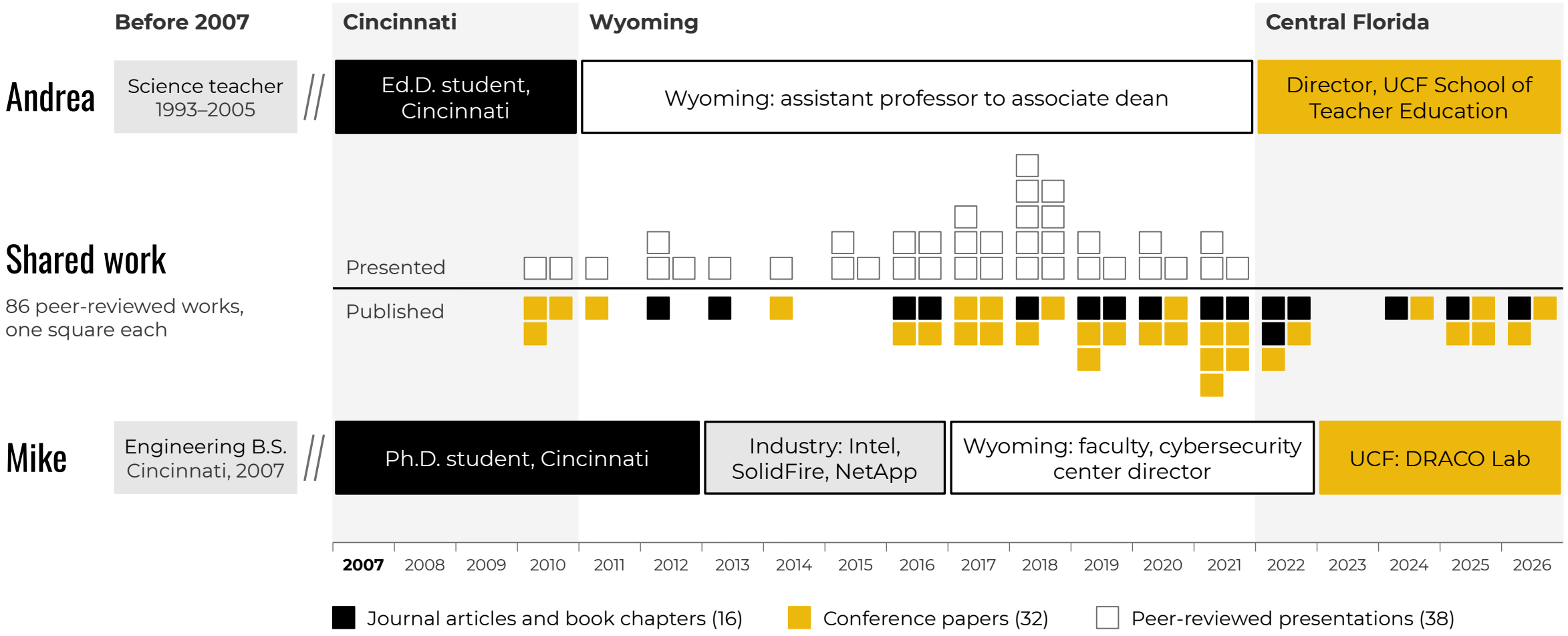
Collaboration and integration across subjects

Andrea C. Borowczak, Ed.D.
Professor and Director, School of Teacher Education

Mike Borowczak, Ph.D.
Associate Professor of Electrical and Computer Engineering · Director, DRACO Lab

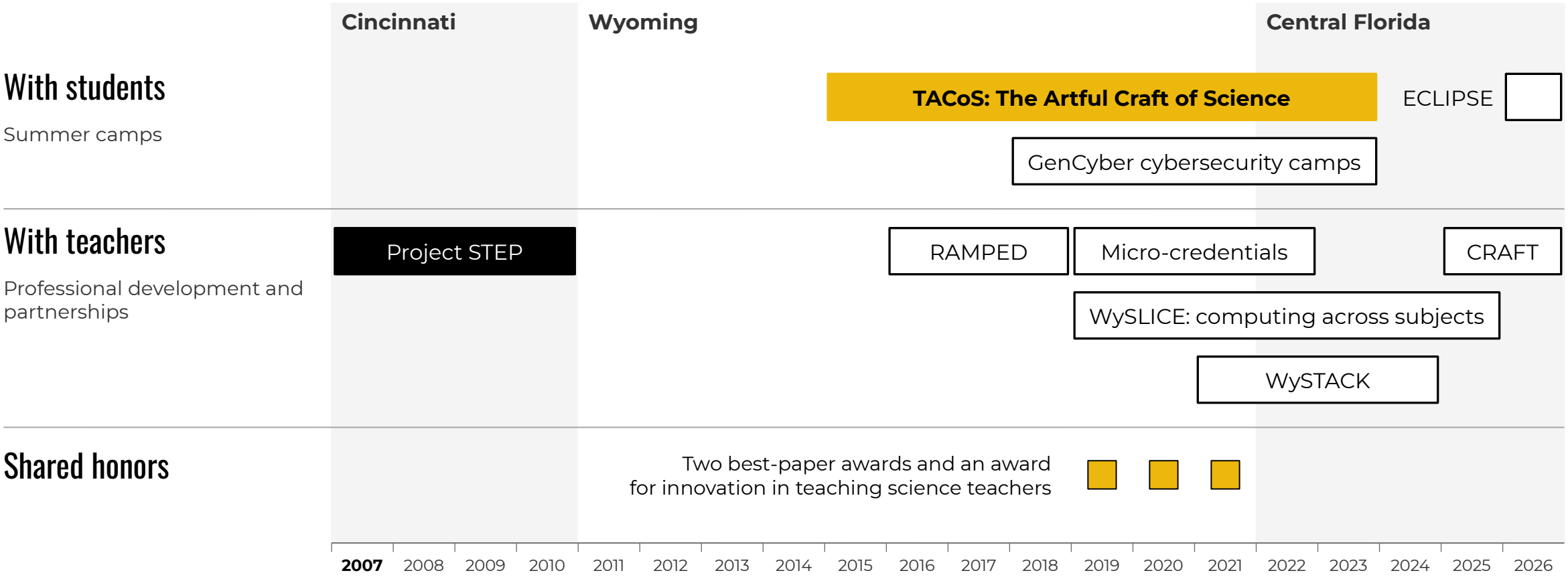
University of Central Florida

Doctoral students in Cincinnati, partners ever since



Source: Mike's curriculum vitae, October 2026

What the partnership built



Funders include the National Science Foundation, the National Security Agency, the Wyoming Department of Education, MilliporeSigma, and Wyoming EPSCoR

One message: collaboration and integration across subjects

1

Pathways into research

Puzzles where subjects meet

2

Publishable work: two editors' views

Readers from both subjects

3

Leading journals and organizations

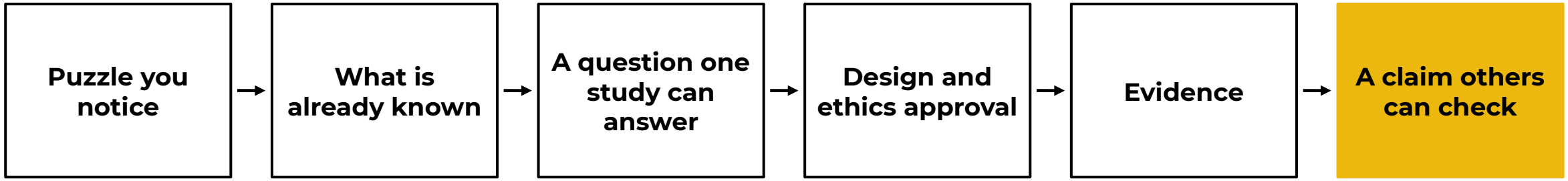
Service in each other's fields

- Both of us in every part · questions: fifteen minutes at the end

Part 1 · Andrea and Mike

Research questions start as puzzles where subjects meet: in a classroom, a studio, or a lab.

From puzzle to publishable question



Often where two subjects meet

Read in both subjects

Narrow until it fits

Plan consent early

Collect what answers the question

State it so a reader could test it

One question, three cohorts

The question

Can educators outside STEM integrate computer science into their own lessons?

- 125 teachers and librarians, kindergarten through grade 12, took part in a virtual integrated STEM and computer science professional development program across three cohorts (2020–2022)
- Mixed methods: surveys, lesson plans, and comparisons across disciplines
- Each educator group compared with STEM specialists

WySLICE (NSF Computer Science for All) · manuscript in revision

Build the team the question needs

Practice

Teachers, artists, and librarians who live the problem

Content

Science, engineering, and arts expertise

Methods

Qualitative and quantitative design and analysis

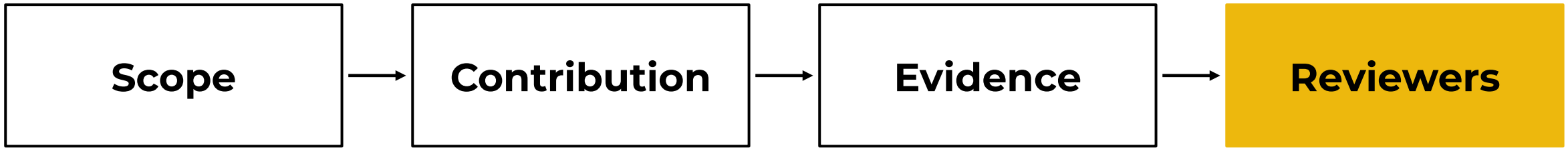
- Our pattern: an engineer and a science educator, with school partners and an external evaluation team

Part 2 · Mike and Andrea

Work across subjects, but write for one journal's readers at a time.

The editor's first pass

Three gates before any reviewer sees the paper



Returned when the paper fits another journal's readers

Returned when no sentence says what the paper adds

Returned when it describes practice with no evidence of what changed

Sent out for peer review

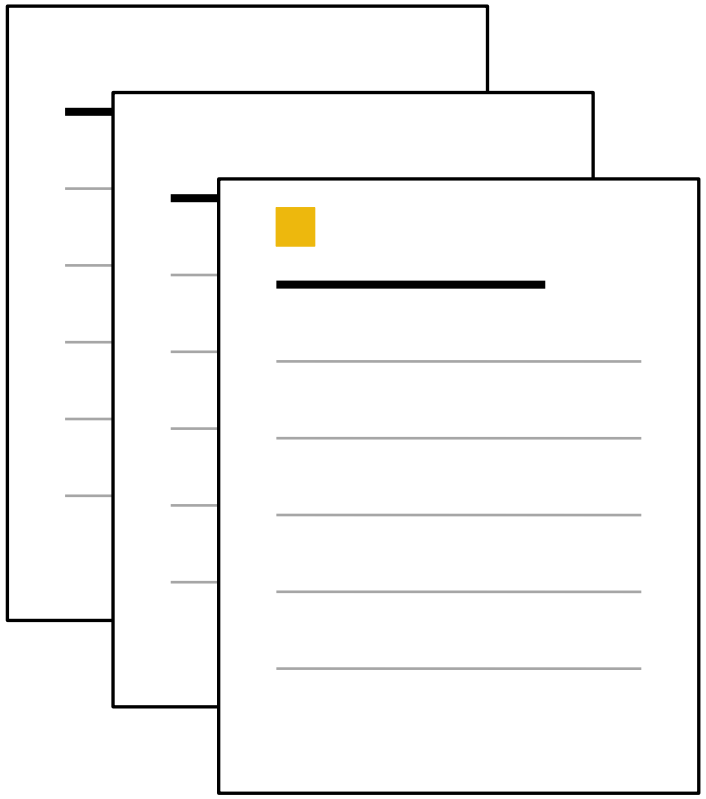
- Write the contribution sentence first

This paper shows _____ for _____, which matters because _____.

- Groups of three, mixed across subjects · three minutes
- Each person drafts one; the group picks the clearest
- Two groups read theirs aloud

Choose the venue before the first draft

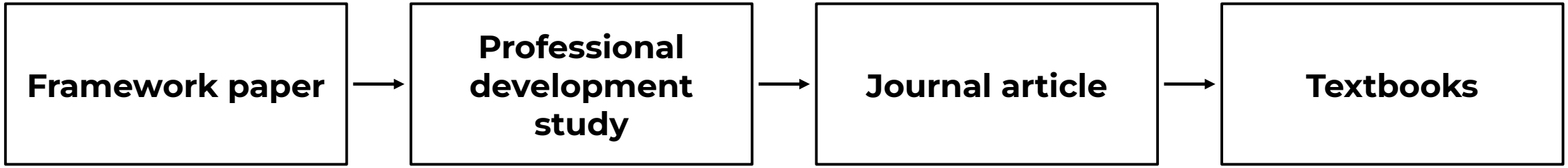
- Read the aims and scope, then three recent papers
- Cite the journal's own conversation
- Match its format: length, structure, and reference style
- Check the review model and the time to a first decision
- Name a second venue in the other subject



Three recent papers

One line of work, several distinct papers

Our CRAFT work, one claim per publication



The idea and its rationale, no data

Teachers' matched pre and post surveys

Framework and evidence together

The framework in daily use

- Test: each paper's contribution sentence differs from the others
- Conference papers can grow into journal articles

Answer reviewers the way engineers verify a design

Reviewer comment	What we changed	Where
1.1 The sample size is unclear	Added a participant table with attrition	Page 6, Table 2
1.2 How valid is the survey?	Cited its validation study and added reliability values	Page 7, lines 210–224
2.1 Who counts as a non-STEM educator?	Defined the groups in the abstract and introduction	Pages 1 and 3

- Answer every comment
- Point to every change
- Disagree with evidence, courteously

Illustrative rows written for this talk

Part 3 · Andrea and Mike

Service across communities shows you how decisions get made.

A ladder you can start this term

5 Lead a journal or an organization

4 Chair a session or guest-edit an issue

3 Volunteer for a committee or working group

2 Join a division or special interest group

1 Co-review one paper a term with your supervisor

Rungs we have climbed

Andrea

- ☐ President, Association for Science Teacher Education (2023)
- ■ Chair, ASEE Pre-College Engineering Education Division (2021–2023)
- ■ Associate editor, Computers in Education (CoED) Journal (six years)

Mike

- ☐ Editor-in-chief, CoED Journal
- ☐ Chair, ASEE Computers in Education Division (2026–)
- ■ Director, Northwest Region, Association for Science Teacher Education (2019–2022)
- ■ Service in the other's professional community

What leading has taught us

Andrea

- Leading a school of teacher education means helping other people's knowledge grow and transfer
- Twenty years in one division: service compounds into community
- [Andrea: a lesson from the ASTE presidency]

Mike

- Every submission shows you the field's full range, from first drafts to finished studies
- A clear aims-and-scope page saves authors months
- Reviewers are volunteers: a specific request gets a faster yes

Five things we would tell our earlier selves

1. Find a collaborator in another subject and keep going: ours is nearly twenty years old
2. Write the contribution sentence before the paper
3. Pick the venue before the first draft
4. Read reviews twice: once for the sting, once for the fix
5. Say yes to service across communities: it shows you how decisions get made

Questions

Thank you to the Arts+STEM Hub at Mary Immaculate College

Andrea C. Borowczak · Andrea.Borowczak@ucf.edu
Mike Borowczak · mike.borowczak@ucf.edu
CoED Journal: coed-journal.org · volunteer reviewers welcome